

The Accessible Color Contrast Checklist (WCAG)

Understanding WCAG Compliance

WCAG = Web Content Accessibility Guidelines

Making your app readable for **everyone**, including:

- People with low vision
 - Color blindness (8% of men, 0.5% of women)
 - Age-related vision changes
 - Users in bright sunlight
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The Golden Ratios

To ensure your app is readable by everyone, meet these **contrast ratios** between text and background:

WCAG AA (Minimum - Legal Standard)

Text Size	Required Ratio
Normal Text (< 18pt or < 14pt bold)	4.5:1
Large Text (≥ 18pt or ≥ 14pt bold)	3:1

WCAG AAA (Enhanced - Recommended)

Text Size	Required Ratio
Normal Text (< 18pt or < 14pt bold)	7:1
Large Text (≥ 18pt or ≥ 14pt bold)	4.5:1

Real-World Examples

Common Failures

Light Grey on White:

None

Background: #FFFFFF (white)

Text: #999999 (light grey)

Ratio: 2.85:1 ← FAILS AA

Blue Link on Blue Background:

None

Background: #2196F3

Text: #1976D2

Ratio: 1.8:1 ← FAILS AA

✓ Passing Examples

Black on White:

None

Background: #FFFFFF

Text: #000000

Ratio: 21:1 ← Passes AAA

Dark Grey on White:

None

Background: #FFFFFF

Text: #595959

Ratio: 7.0:1 ← Passes AAA for normal text

White on Dark Blue:

None

Background: #1565C0

Text: #FFFFFF

Ratio: 8.6:1 ← Passes AAA

Testing Checklist

Text Elements to Check

- ☐ Body text
- ☐ Headings
- ☐ Button text
- ☐ Link text (both default and hover states)
- ☐ Form labels
- ☐ Placeholder text
- ☐ Error messages
- ☐ Disabled state text

UI Elements

- ☐ Icon-only buttons (need text alternatives)
 - ☐ Focus indicators (keyboard navigation)
 - ☐ Active/selected states
 - ☐ Breadcrumbs
 - ☐ Tags/badges
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Quick Tips for Designers

1. Pure Grey Problem

✗ Avoid: Light grey text (#999, #AAA, #BBB) on white backgrounds

Why: Rarely passes AA standard

✓ Solution: Use #595959 or darker for normal text on white

2. Button Contrast

Buttons need **TWO** passing contrasts:

1. Button text vs. button background
2. Button background vs. page background

Example:

None

Page: White (#FFFFFF)

Button: Blue (#2196F3)

Text: White (#FFFFFF)

Check 1: White text on Blue button = 4.5:1 

Check 2: Blue button on White page = 3.1:1  (for large UI component)

3. Color Blindness Considerations

Never use color ALONE to convey meaning:

 **Bad:**

- Red text for errors (colorblind users won't see it)
- Green/Red for status without labels

 **Good:**

-  Icon + Red text for errors
-  Icon + Green text for success
- "Error:" label + red color

Types of Color Blindness:

- **Protanopia:** Can't see red
- **Deuteranopia:** Can't see green
- **Tritanopia:** Can't see blue
- **Achromatopsia:** No color vision (rare)

4. Link Visibility

Links must be distinguishable from body text without relying on color alone.

Options:

- Underline
- Bold weight
- Icon
- Different font

Note: 3:1 contrast ratio between link and surrounding text if no non-color cue exists.

Contrast Ratio Calculation

Formula:

None

$$(L1 + 0.05) / (L2 + 0.05)$$

Where:

L1 = Relative luminance of lighter color

L2 = Relative luminance of darker color

Relative Luminance Formula:

None

$$L = 0.2126 * R + 0.7152 * G + 0.0722 * B$$

Where R, G, B are gamma-corrected values

💡 **Don't calculate manually** — use tools like:

- WebAIM Contrast Checker
- Colour Contrast Analyser
- Browser DevTools
- Our Color Picker Tool

Color Palette Best Practices

Building Accessible Palettes

Start with extremes:

1. Choose your brand color
2. Create a dark version for text (test contrast)
3. Create a light version for backgrounds

Example Palette:

None

Primary: #1976D2 (Blue)

└ Dark (for text on light): #0D47A1 (8.2:1 on white) ✓

└ Medium (for buttons): #1976D2 (4.6:1 on white) ✓

└ Light (for backgrounds): #E3F2FD (1.1:1 with white)

Text on #E3F2FD background: Use #0D47A1 (7.5:1) ✓

Safe Color Combinations

Background	Safe Text Colors	Contrast
#FFFFFF (White)	#000000 (Black)	21:1
#FFFFFF (White)	#333333 (Dark Grey)	12.6:1
#FFFFFF (White)	#595959 (Medium Grey)	7.0:1
#000000 (Black)	#FFFFFF (White)	21:1
#F5F5F5 (Off-White)	#333333	11.8:1
#1976D2 (Blue)	#FFFFFF	4.6:1

Testing Tools

Browser Extensions

- **WAVE** (Chrome/Firefox) - Page-wide analysis
- **axe DevTools** - Detailed accessibility audit
- **Lighthouse** (Chrome DevTools) - Built-in scoring

Online Tools

- **WebAIM Contrast Checker** - Quick ratio check
- **Coolors Contrast Checker** - Palette testing
- **Accessible Colors** - Suggests fixes

Design Tools

- **Figma**: Stark plugin, A11y - Color Contrast Checker
 - **Adobe XD**: Color Contrast Analyzer plugin
 - **Sketch**: Contrast plugin
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Common Mistakes

✗ Mistake 1: Only Testing Desktop

Mobile screens in sunlight reduce perceived contrast by 50%

Solution: Test in bright conditions or aim for AAA

✗ Mistake 2: Forgetting Hover/Focus States

```
CSS
a {
  color: #1976D2; /* 4.6:1 ✓ */
}
a:hover {
  color: #64B5F6; /* 2.3:1 ✗ FAILS */
}
```

✗ Mistake 3: Transparent Overlays

```
CSS
.overlay {
  background: rgba(0, 0, 0, 0.5); /* Contrast depends on image */
  color: white; /* May fail */
}
```

Solution: Add solid background or strong text shadow

✗ Mistake 4: Brand Colors Trump Accessibility

"Our brand blue doesn't pass contrast" → Adjust it. Brand recognition < Legal compliance & usability

Legal Requirements

Who Must Comply?

Required by Law:

- U.S. Government websites (Section 508)
- E-commerce sites (ADA Title III)
- Educational institutions
- Financial services
- Healthcare providers

Best Practice for All: Even if not legally required, accessibility:

- Increases user base
- Improves SEO
- Reduces support tickets
- Shows corporate responsibility

Recent Lawsuits

- Domino's Pizza (2019) - \$4,000+ damages
 - Beyoncé's website (2019) - Settlement
 - Hundreds of ADA Title III suits filed yearly
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Implementation Checklist

Development Phase

- ☐ Define color system with contrast-tested values
- ☐ Create CSS variables for approved colors
- ☐ Document approved combinations
- ☐ Set up automated testing (e.g., axe-core, pa11y)

Design Phase

- ☐ Test all mockups with contrast checker
- ☐ Provide text alternatives for all visual info
- ☐ Don't rely on color alone for state changes
- ☐ Document color usage guidelines

QA Phase

- ☐ Manual testing with screen reader
 - ☐ Automated accessibility scan
 - ☐ Test with color blindness simulator
 - ☐ Test on mobile in bright light
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Accessible Color System Template

CSS

```
:root {  
  /* Text on Light Backgrounds */  
  --text-primary: #212121;    /* 16.1:1 on white ✓ */  
  --text-secondary: #666666;  /* 5.7:1 on white ✓ */  
  --text-disabled: #9E9E9E;   /* 2.8:1 (decorative only) */  
  
  /* Text on Dark Backgrounds */  
  --text-on-dark: #FFFFFF;     /* 21:1 on black ✓ */  
  --text-on-dark-secondary: #B0B0B0; /* 8.6:1 on black ✓ */
```

```
/* Interactive Elements */
--link-color: #0D47A1;      /* 8.2:1 on white ✓ */
--link-hover: #1565C0;     /* 6.5:1 on white ✓ */
--focus-outline: #2962FF; /* 5.9:1 on white ✓ */

/* Status Colors */
--success: #2E7D32;        /* 5.4:1 on white ✓ */
--error: #C62828;          /* 5.9:1 on white ✓ */
--warning: #F57C00;         /* 3.9:1 on white, use bold ✓ */

/* Backgrounds */
--bg-primary: #FFFFFF;
--bg-secondary: #F5F5F5;
--bg-dark: #212121;
}
```

Need a Professional UI Audit?

Devxhub builds beautiful, accessible interfaces

Our design team ensures:

- ✓ WCAG 2.1 AA/AAA compliance
- ✓ Color-blind friendly palettes
- ✓ Screen reader optimization
- ✓ Keyboard navigation
- ✓ Mobile-responsive design

Services:

- Accessibility audit & remediation
- Design system creation
- WCAG compliance consulting
- User testing with diverse abilities

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